GENOMEDL CLINICAL GENOMICS PIPELINE (v3)

Pediatric & Germline Hereditary Cancer Screening Report

Sample Identifier:	HG002_CANCER_PANEL.VCF	**Reference Genome:**	GRCh38 / hg38
Panel Profile:	130 Hereditary Cancer Genes	**Analysis Date:**	November 13, 2026 (08:13 UTC)
Total Variants Filtered:	12044 candidates	**Actionable Tiers:**	**0 Pathogenic / Likely Pathogenic**

NO HIGH-PENETRANCE PATHOGENIC VARIANTS IDENTIFIED

No pathogenic or likely pathogenic variants were detected in the 130 evaluated cancer predisposition genes.

2. Detailed Genomic Variant Classifications

Each candidate variant was filtered against gnomAD population frequencies (AF > 1% = Benign), matched against ClinVar curated submissions (≥2 stars), and unclassified novel variants were scored with the GenomeDL v3 ESM-augmented Multimodal Fusion ResNet.

GENE & LOCUS	HGVS / PROTEIN	GT (DP)	CLASSIFICATION	EVIDENCE SOURCE	MODEL P
SDHB 1:17016703 C>T	intronic	0/1 (Het, DP=1058)	**Benign**	GenomeDL v3 ESM-Augmented Fusion ResNet	0.003
SDHB 1:17017409 A>C	intronic	0/1 (Het, DP=741)	**Benign**	GenomeDL v3 ESM-Augmented Fusion ResNet	0.004
SDHB 1:17017581 A>G	intronic	0/1 (Het, DP=468)	**Benign**	GenomeDL v3 ESM-Augmented Fusion ResNet	0.002
SDHB 1:17017653 G>A	intronic	0/1 (Het, DP=754)	**Benign**	GenomeDL v3 ESM-Augmented Fusion ResNet	0.002
SDHB 1:17018578 G>GC	intronic	1/1 (Hom, DP=890)	**Benign**	GenomeDL v3 ESM-Augmented Fusion ResNet	0.002
SDHB 1:17019410 A>G	intronic	0/1 (Het, DP=1187)	**Benign**	GenomeDL v3 ESM-Augmented Fusion ResNet	0.002
SDHB 1:17019909 A>T	intronic	1/1 (Hom, DP=1015)	**Benign**	GenomeDL v3 ESM-Augmented Fusion ResNet	0.003
SDHB 1:17021461 G>A	intronic	1/1 (Hom, DP=1040)	**Benign**	GenomeDL v3 ESM-Augmented Fusion ResNet	0.002
SDHB 1:17022049 T>G	intronic	0/1 (Het, DP=902)	**Benign**	GenomeDL v3 ESM-Augmented Fusion ResNet	0.002
SDHB 1:17022192 C>T	intronic	0/1 (Het, DP=898)	**Benign**	GenomeDL v3 ESM-Augmented Fusion ResNet	0.003

**Note: 12,034 additional benign polymorphisms are omitted from this summary table for clinical conciseness. Full records are archived in the structured JSON report.*

4. Computational Methodology & Validation Standards

Model Architecture: Multimodal Fusion ResNet combining 26 tabular biochemical features, 8-d consequence/syndrome embeddings, and ESM-2 (t12_35M_UR50D) masked-marginal log-likelihood ratios and 480-d hidden states.
Calibration & Benchmark: Validation-fitted Platt logit scaling (scale=1.0559, bias=-1.5561; ECE 0.0279 → 0.0009; checkpoint test ROC-AUC 0.9929). On 5-fold Gene-Held-Out cross-validation (GroupKFold), Fusion ResNet achieves overall ROC-AUC **0.9919 [95% CI: 0.9913 – 0.9925]** vs. HistGradientBoosting baseline 0.9621 (missense subset: 0.8857 vs. 0.8976).

MANDATORY CLINICAL GENOMICS DISCLAIMER:
Research prototype, not a diagnosis. Confirm in a certified clinical diagnostic laboratory with formal genetic counselling. Testing children or minors for adult-onset conditions requires specialized pediatric genetic counselling and informed consent.